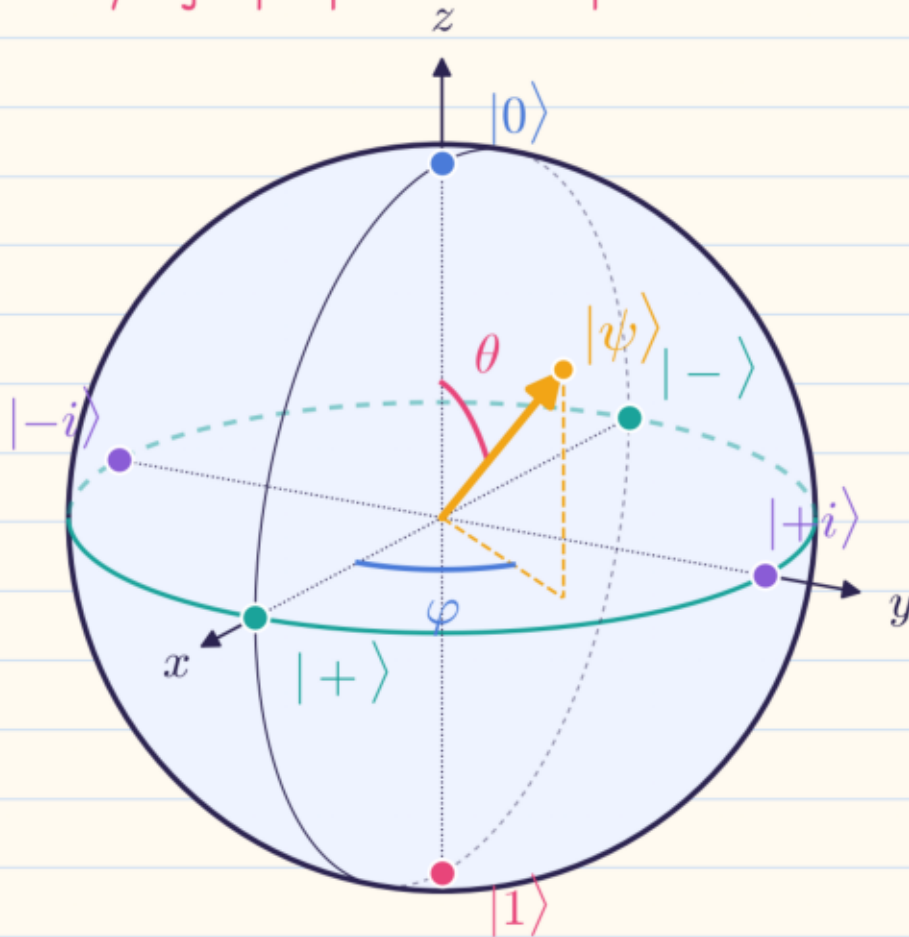


Meet the Bloch sphere

every single-qubit pure state is one point on this ball



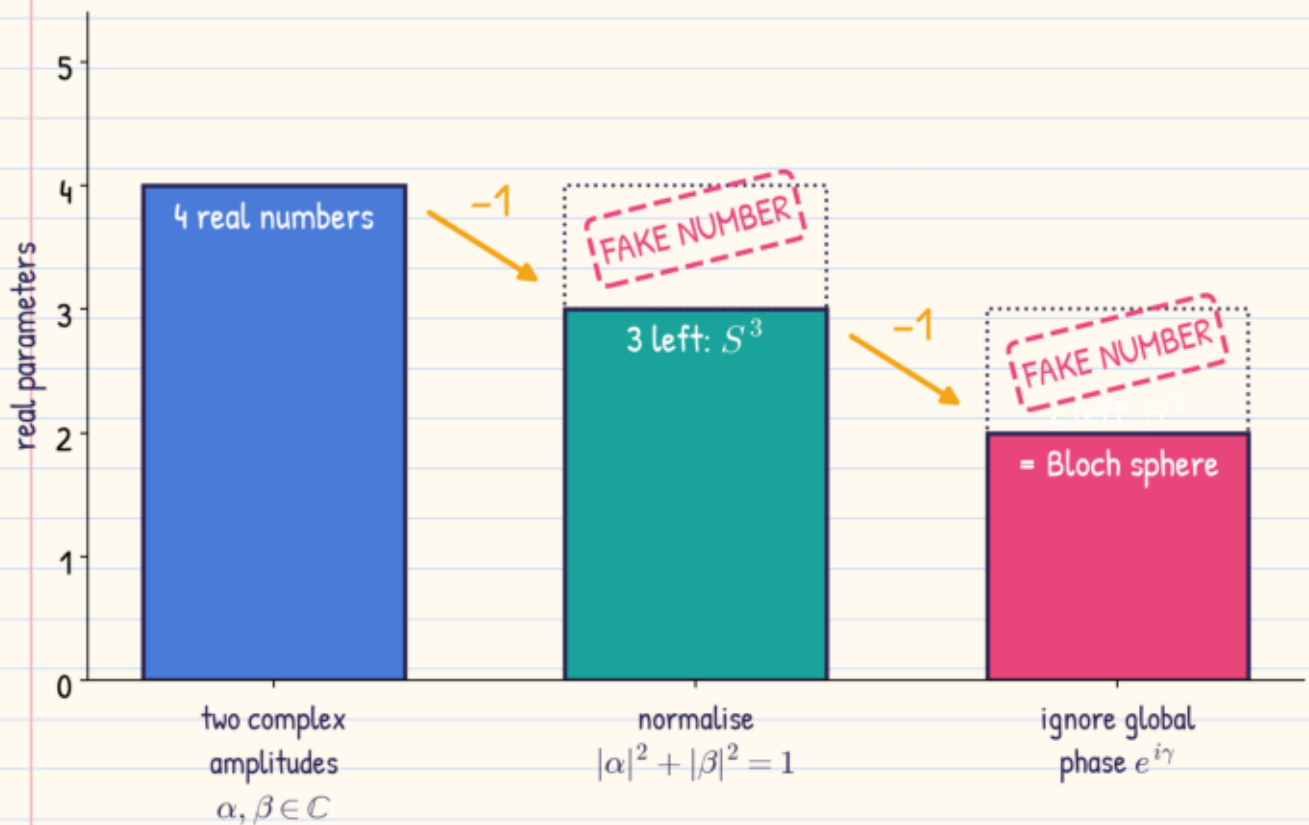
$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle$$

$0 \leq \theta \leq \pi$ (down from the north pole) $0 \leq \varphi < 2\pi$ (round the equator from x)

The Bloch sphere. $|0\rangle$ sits at the north pole, $|1\rangle$ at the south pole, and the four equal superpositions sit round the equator. Any pure state is an arrow given by two angles: θ measured down from the north pole and φ measured round the equator.

The great parameter countdown

why four numbers become a two-dimensional sphere

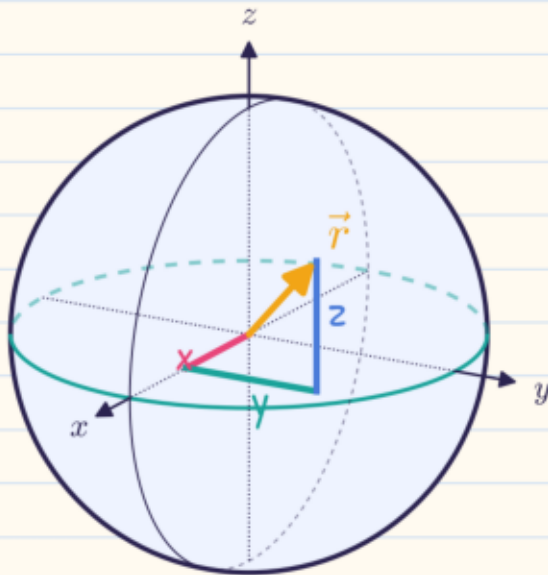


Four is too many: only two angles, θ and φ , actually matter.

Two complex amplitudes look like 4 real numbers. Normalisation knocks off one (leaving the 3-sphere S^3), and the unobservable global phase knocks off another, leaving 2: the surface of the Bloch sphere.

Where exactly is the arrow?

Bloch vector coordinates



from the angles

$$x = \sin \theta \cos \varphi$$

$$y = \sin \theta \sin \varphi$$

$$z = \cos \theta$$

from the amplitudes

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$x = 2 \operatorname{Re}(\alpha^* \beta)$$

$$y = 2 \operatorname{Im}(\alpha^* \beta)$$

$$z = |\alpha|^2 - |\beta|^2$$

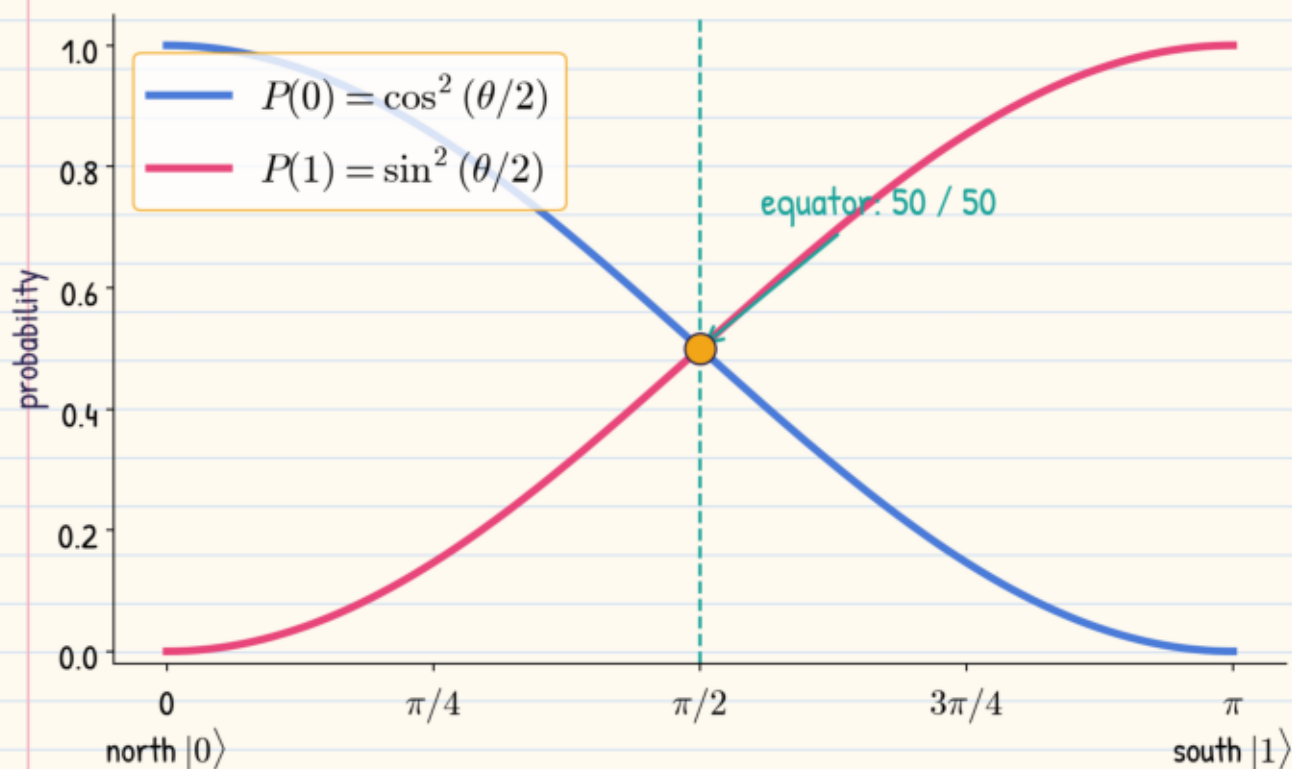
and $x^2 + y^2 + z^2 = 1$ for a pure state

(each is an average: $x = \langle \sigma_x \rangle$ etc.)

The arrow's (x, y, z) coordinates, worked out either from the angles θ and φ or straight from the amplitudes α and β . Each coordinate is also the average result of measuring the matching Pauli operator.

Measurement odds

how likely are you to get 0 or 1?



φ doesn't change these odds at all, and $P(0) + P(1) = 1$ always.

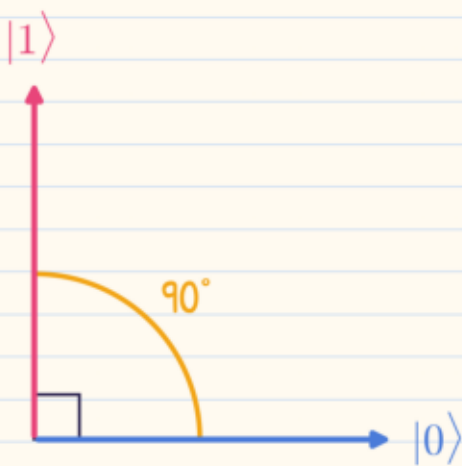
The chance of measuring 0 or 1 depends only on the latitude θ . At the north pole you always get 0, at the south pole always 1, and anywhere on the equator it's a fair coin.

Why the sneaky $\Theta/2$?

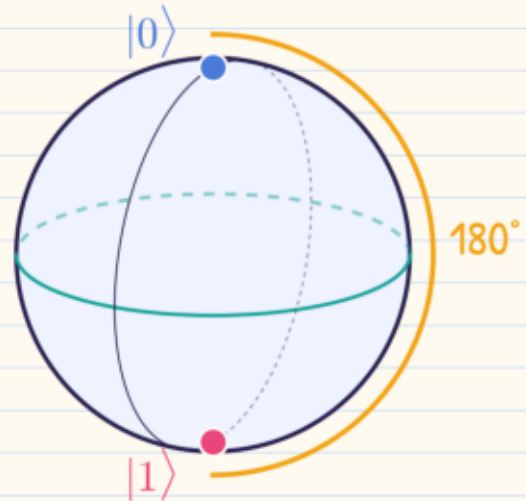
angles double on the way to the sphere

Hilbert space: orthogonal

Bloch sphere: opposite poles



$\times 2$



overlap: $|\langle\psi|\chi\rangle|^2 = \cos^2(\Theta/2)$, Θ = angle between them on the sphere

States that are at right angles in Hilbert space (zero overlap) end up at opposite ends of the sphere, so every angle doubles. The overlap of two states is $\cos^2(\Theta/2)$, where Θ is their angle apart on the sphere:

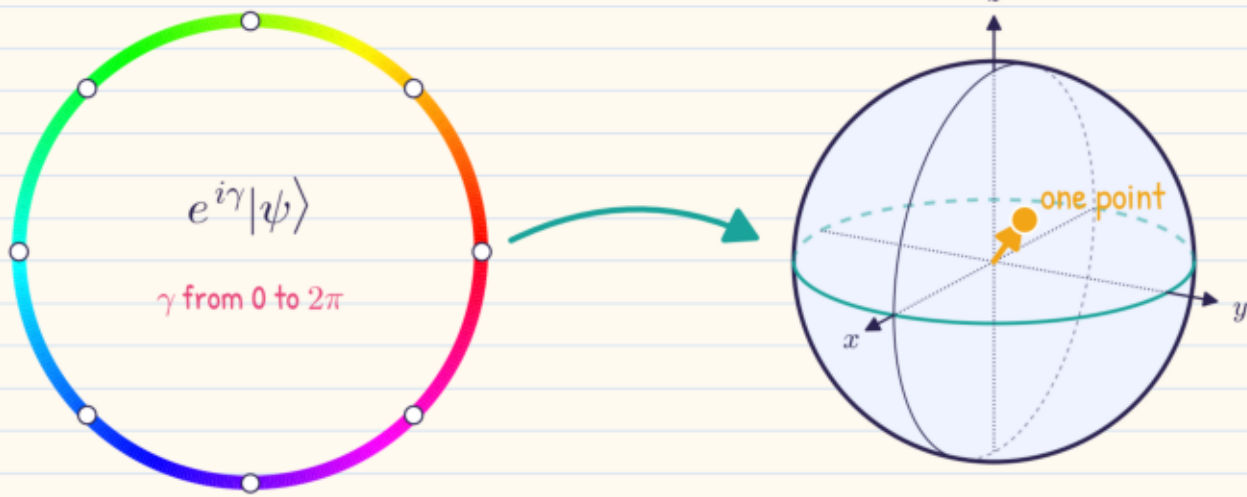
0° gives 1, 180° gives 0.

Global phase does nothing

a whole circle of phases lands on one point

the phase circle (a fibre)

the Bloch sphere

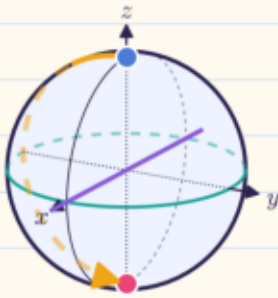


$$|\langle \phi | e^{i\gamma} \psi \rangle|^2 = |\langle \phi | \psi \rangle|^2: \text{no measurement can tell them apart (the Hopf map } S^3 \rightarrow S^2)$$

Multiplying a state by an overall phase $e^{i\gamma}$ changes no measurement probabilities. So a whole circle of states (one 'fibre' of the Hopf fibration from S^3 to S^2) all land on the same single point of the sphere.

Gates are rotations

each single-qubit gate turns the whole sphere

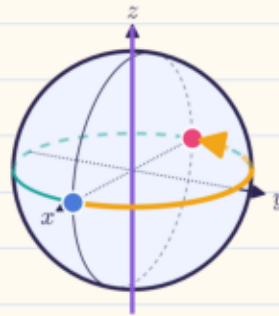


X

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

π about x

bit flip: $|0\rangle \rightarrow |1\rangle$

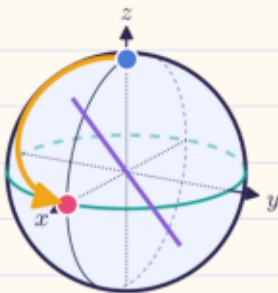


Z

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

π about z

phase flip: $|+\rangle \rightarrow |-\rangle$

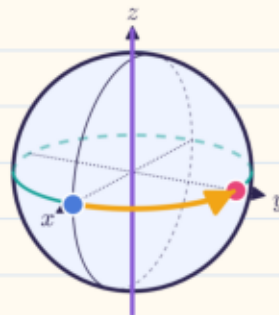


H

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

π about $(x+z)/\sqrt{2}$

swaps $z \leftrightarrow x$: $|0\rangle \rightarrow |+\rangle$



S

$$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

$\pi/2$ about z

quarter turn: $|+\rangle \rightarrow |i\rangle$

Single-qubit gates are rotations of the sphere. X turns it half-way round the x axis (0 becomes 1), Z half-way round z, H half-way round the diagonal between x and z (so it swaps those axes), and S a quarter turn round z. Purple is the rotation axis, blue the start, pink the end.

a pulse of light or microwaves steers the arrow

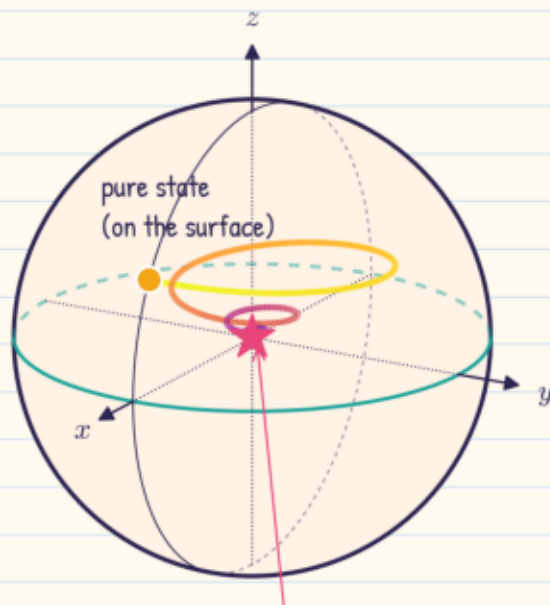
e.g. $R_x(\pi) = -iX$ (same as X up to a global phase)



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Inside the ball: mixed states

when the qubit loses its nerve



centre: maximally mixed
 $\rho = I/2$

$$\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma})$$

$$|\vec{r}| \leq 1$$

$|\vec{r}| = 1$: pure, on the surface

$|\vec{r}| < 1$: mixed, inside

$\vec{r} = 0$: coin toss, $\rho = I/2$

T_2 (dephasing): x, y shrink

T_1 (relaxation): z heads to its
resting value

(here drawn spiralling to the centre; real T_1 decay
pulls towards $|0\rangle$, the ground state)

Noisy qubits are described by a density matrix, whose Bloch vector $\vec{r}$ can point inside the ball. Pure states live on the surface, the completely random state sits at the centre, and decoherence (T_2 shrinking x and y , T_1 relaxing z) drags the arrow inwards.

The Paulis and their six eigenstates

the six cardinal points of the sphere

Pauli	matrix	eigenvalue +1	eigenvalue -1	axis
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$\sigma_x = X$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$ +\rangle = \frac{ 0\rangle + 1\rangle}{\sqrt{2}}$	$ -\rangle = \frac{ 0\rangle - 1\rangle}{\sqrt{2}}$	$\pm x$
----------------	--	--	--	---------

$\sigma_y = Y$	$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$	$ +i\rangle = \frac{ 0\rangle + i 1\rangle}{\sqrt{2}}$	$ -i\rangle = \frac{ 0\rangle - i 1\rangle}{\sqrt{2}}$	$\pm y$
----------------	---	--	--	---------

$\sigma_z = Z$	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	$ 0\rangle$	$ 1\rangle$	$\pm z$
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$$XY = iZ, YZ = iX, ZX = iY, X^2 = Y^2 = Z^2 = I$$

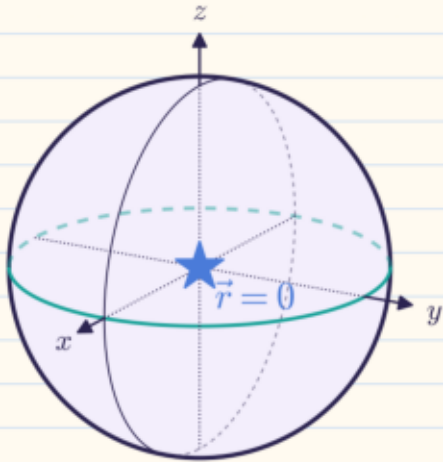
The three Pauli matrices. Each has two eigenstates, and those six states are exactly the six points where the x, y and z axes poke through the sphere. The +1 eigenstate is on the positive end of the axis.

Two qubits: the picture breaks

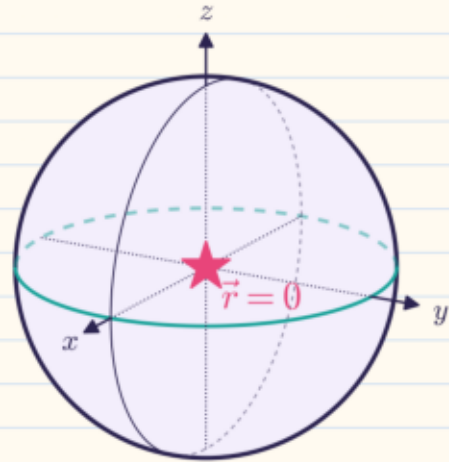
a Bell state can't be drawn as two points

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

qubit A: $\rho = I/2$



qubit B: $\rho = I/2$



Each half on its own is perfectly random; all the information hides in the correlations.

For an entangled Bell state, each qubit on its own is maximally mixed:
both Bloch vectors shrink to zero length at the centre. The
interesting stuff lives in the correlations between them, which one
sphere per qubit can't show.

Bloch sphere cheat sheet

stick this on the fridge

state

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle$$

Bloch vector

$$\vec{r} = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta)$$

from amplitudes

$$x = 2\text{Re}(\alpha^* \beta), \quad y = 2\text{Im}(\alpha^* \beta), \quad z = |\alpha|^2 - |\beta|^2$$

probabilities

$$P(0) = \cos^2 \frac{\theta}{2} = \frac{1+z}{2}, \quad P(1) = \sin^2 \frac{\theta}{2} = \frac{1-z}{2}$$

overlap

$$|\langle \psi | \chi \rangle|^2 = \cos^2 \frac{\Theta}{2} = \frac{1 + \vec{r}_\psi \cdot \vec{r}_\chi}{2}$$

parameters

$$4 \rightarrow 3 (S^3) \rightarrow 2 (S^2)$$

rotation

$$R_{\hat{n}}(\alpha) = \cos \frac{\alpha}{2} I - i \sin \frac{\alpha}{2} \hat{n} \cdot \vec{\sigma}$$

gates

$$X: \pi \text{ about } x; \quad Z: \pi \text{ about } z; \quad S: \frac{\pi}{2} \text{ about } z; \quad H: \pi \text{ about } \frac{x+z}{\sqrt{2}}$$

Rabi

$$P(1) = \sin^2 (\Omega t / 2)$$

mixed states

$$\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma}), \quad |\vec{r}| \leq 1$$

Every key formula on one page: the state, its coordinates, the odds, overlaps, the parameter countdown, rotations and gates, Rabi oscillation and mixed states.